STRUCTURE OF THE NORMALIZATION OF A LOCALLY NOETHERIAN SCHEME OF DIMENSION ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- General properties of the normalization morphism: reducedness, stalks, normality, birationality. Stacks Project, Morphisms, Lemmas morphisms-lemma-normalization-reduced (tag 035L), morphisms-lemma-stalk-normalization (tag 035Q), morphisms-lemma-normalization-normal (tag 035N) and morphisms-lemma-normalization-birational (tag 035P). Dispose of the parts of the statement that hold for every normalization.
- Krull-Akizuki in the form of the integral closure of a one-dimensional Noetherian domain being Dedekind. Stacks Project, Algebra, Lemma algebra-lemma-integral-closure-Dedekind (tag 034W). Provides finiteness of fibres and residue field extensions and the local Dedekind description.
- Description of the normalization used for the affine reduction. Stacks Project, Morphisms, Lemma morphisms-lemma-description-normalization (tag 035M). Reduces the remaining properties to the one-dimensional Noetherian domain case.

Source reference. Original reference 035O: sources/stacks-chow/tags.txt, source lines 4135–4135. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C3B: sources/stacks-chow/tags.txt, source lines 15186–15186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 035Q: sources/stacks-chow/tags.txt, source lines 4137–4137. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BXC: sources/stacks-chow/tags.txt, source lines 14975–14975. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 09IG: sources/stacks-chow/tags.txt, source lines 12039–12039. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a locally Noetherian scheme of dimension 1. Let $\nu : X^\nu \to X$ be the normalization. Then*

(1) *ν is integral, surjective, and induces a bijection on irreducible components,*
(2) *there is a factorization $X^\nu \to X_{red} \to X$ and the morphism $X^\nu \to X_{red}$ is the normalization of X_{red},*
(3) *$X^\nu \to X_{red}$ is birational,*
(4) *for every closed point $x \in X$ the stalk $(\nu_*\mathcal{O}_{X^\nu})_x$ is the integral closure of $\mathcal{O}_{X,x}$ in the total ring of fractions of $(\mathcal{O}_{X,x})_{red} = \mathcal{O}_{X_{red},x}$,*
(5) *the fibres of ν are finite and the residue field extensions are finite,*
(6) *X^ν is a disjoint union of integral normal locally Noetherian schemes and each affine open is the spectrum of a finite product of Dedekind domains.*

Proof. Many of the results are in fact general properties of the normalization morphism, see Morphisms, Lemmas 035O, 0C3B, 035Q, and 0BXC. What is not clear is that the fibres are finite, that the induced residue field extensions are finite, and that X^ν locally looks like the spectrum of a Dedekind domain (and hence is Noetherian). To see this we may assume that $X = \mathrm{Spec}(A)$ is affine, Noetherian, dimension 1, and that A is reduced. Then we may use the description in Morphisms, Lemma 035P to reduce to the case where A is a Noetherian domain of dimension 1. In this case the desired properties follow from Krull-Akizuki in the form stated in Algebra, Lemma 09IG. □